ANGUS BARLOW

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ML RESEARCH ENGINEER | LLM POST-TRAINING, FINE-TUNING & EVALUATION

PROFESSIONAL SUMMARY

Computer Science graduate (B.S. CS, Magna Cum Laude, 3.702 GPA) pursuing an M.S. at NC State, with an Applied AI internship at KBI Biopharma. Builds post-training and evaluation tooling for language models. Recent work fine-tuned a 30B open-weight model with attention LoRA and implemented DPO/GRPO from scratch, measured with behavioral batteries rather than train metrics.

TECHNICAL SKILLS

Post-Training: PyTorch, TRL, PEFT, attention LoRA, SFT, DPO, GRPO, reward shaping, KL-constrained RL, GGUF, Ollama
Evaluation: behavioral test batteries, held-out entity splits, edge-case scoring suites, raw-generation auditing, human-in-the-loop review
Languages & Data: Python, SQL, Bash, Git, GitHub Actions CI/CD, pandas, NumPy, FastAPI, Docker, pytest

EXPERIENCE

KBI Biopharma | Applied AI Intern | 2026

- Built a Python and FastAPI prototype that uses AI to review biopharmaceutical deviation investigations, converting heterogeneous investigation records into typed schemas linked to their source records.
- Wrote an evaluation suite that scores review outputs against documented edge cases before stakeholders see them.
- Designed a fine-tuning program for the review models using reviewer approve/deny decisions with written reasons as the training signal. Evaluated closed-API and open-weight model paths and raised the safety and data-use concerns that paused the effort before any training run.
- Automated data collection from Microsoft Graph and SharePoint, with paging and recovery after failed runs.

CareYaya, Boston, MA | Software Engineer & Data Scientist | 2024 - 2026

- Built backend services and data pipelines in Python for a caregiver-marketplace platform connecting families with student caregivers.
- Ran SQL and Python analyses over platform and care-delivery data that fed matching and operations decisions.
- Prepared training and evaluation data for the company's applied NLP work in elder care.

SELECTED TECHNICAL PROJECTS

llm-posttraining - Post-Training Research Codebase

Python | PyTorch | TRL | DPO | GRPO | github.com/barlowa124/llm-posttraining

- Implemented GRPO over k=6 rollouts with KL to a frozen reference and DPO from the objective on SmolLM2-135M, with every stage measured on held-out entities and raw generations committed.
- Diagnosed a DPO collapse that produced 95% degenerate held-out outputs at 1.0 train preference accuracy, then repaired it with a shaped reward tier that restored abstention from 5% to 100% in 8 gradient steps, replicated across rollout seeds.

qwen3-coder-30b-agent-lora - Local-Agent Tool-Call Fine-Tune

Python | TRL | PEFT | attention LoRA | GGUF | Ollama | github.com/barlowa124

- Fine-tuned Qwen3-Coder-30B-A3B with attention LoRA (r=16 on q/k/v/o projections, NF4-quantized base, TRL SFTTrainer and PEFT) over 189 local-agent trajectories across multiple rounds, merged to GGUF and served in Ollama.
- Evaluated adapter rounds with a behavioral battery that executes real tool calls through a chat-shim API against the base model.

trust-tools - Agent Trace Auditing & Policy Checking

Python | deterministic evaluation | github.com/barlowa124/trust-tools

- Built tools for auditing agent traces and checking tool calls against policy rules.
- Implemented a deterministic evaluation runner and hash-chained evidence records, including adversarial examples that expose limits of the checks.

EDUCATION

North Carolina State University | Master of Biomanufacturing (In Progress, Expected January 2028) | Raleigh, NC
Southern New Hampshire University | B.S. Computer Science & B.S. Environmental Science | Jan 2026
Magna Cum Laude | GPA 3.702 | Coursework: Data Structures & Algorithms, Software Security, Software Testing/Automation QA, Client/Server Development